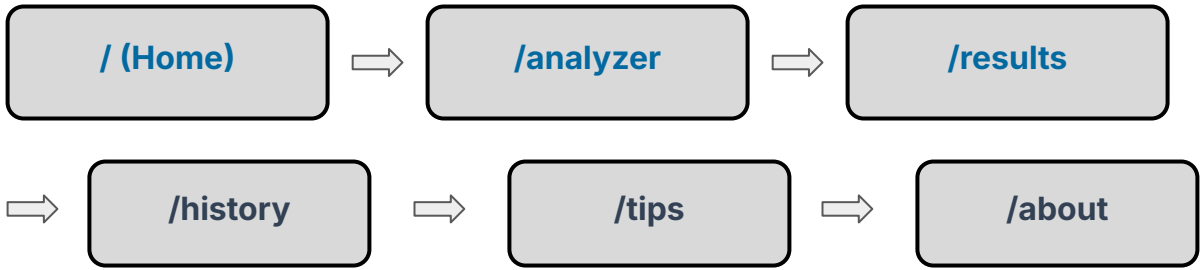
Site Structure & Route Flow Diagram



State Dataclass

current_password: (str): Stores user input text entered in the main analyzer input field.

history (list[PasswordRecord]): Persistent list storing instances of past analyzed passwords.

strict_mode (bool): Toggles strict validation rules (e.g., symbol enforcement).

active_category (str): Selected target rules to check the password strength (e.g., "Personal", "Corporate", "Government").

PasswordRecord Dataclass

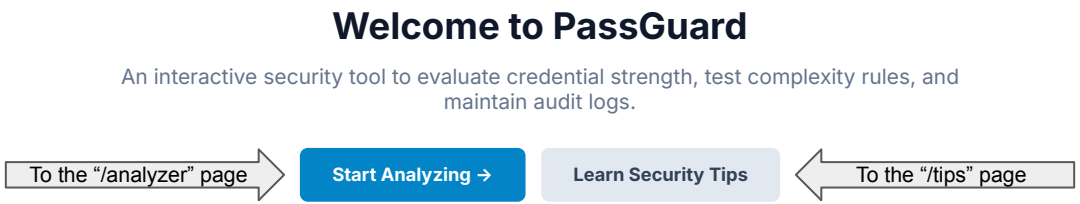
masked_password (str): Partially obscured version of the password for privacy with “***” (e.g., P@ss****).

score (int): Security rating calculated during analysis (0 to 100).

rating_label (str): Summary evaluation string (e.g., "Weak", "Moderate", "Strong").

Page 1: Home Page (“/”)

Parameters: state: State

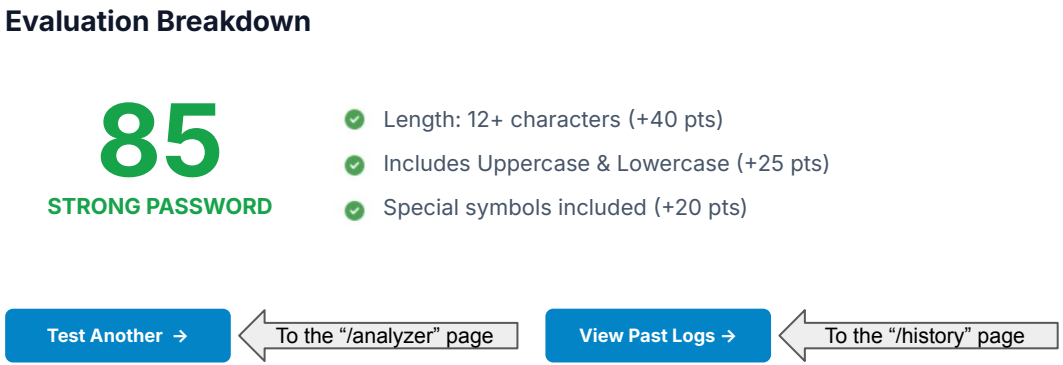


Page 2: Analyzer (“/analyzer”)



Page 3: Results (“/results”)

Parameters: state: State



Page 4: History (“/history”)

Parameters: state: State

LOOPS OVER STATE.HISTORY

P@ss****2026 — Score: 85 (Strong)

admin**** — Score: 20 (Weak)

Return Home → — To the “/home” page

Page 5: Tips & Best Practices (“/tips”)

Parameters: state: State



Use Passphrases

Combine 4+ random words to create memorable, high-entropy secrets.



Password Manager

Never reuse credentials across multiple critical online services.



Enable 2FA

Add an extra layer of defense beyond traditional password strings.

Return Home →

To the “/home” page

Page 6: About (“/about”)

Parameters: state: State

About & Methodology



Entropy Calculation

Explains how mathematical pool sizing and character length dictate credential security score.



Policy Standards

Outlines the rules used for Personal, Corporate, and Government category evaluations.

Return Home →

To the “/home” page